

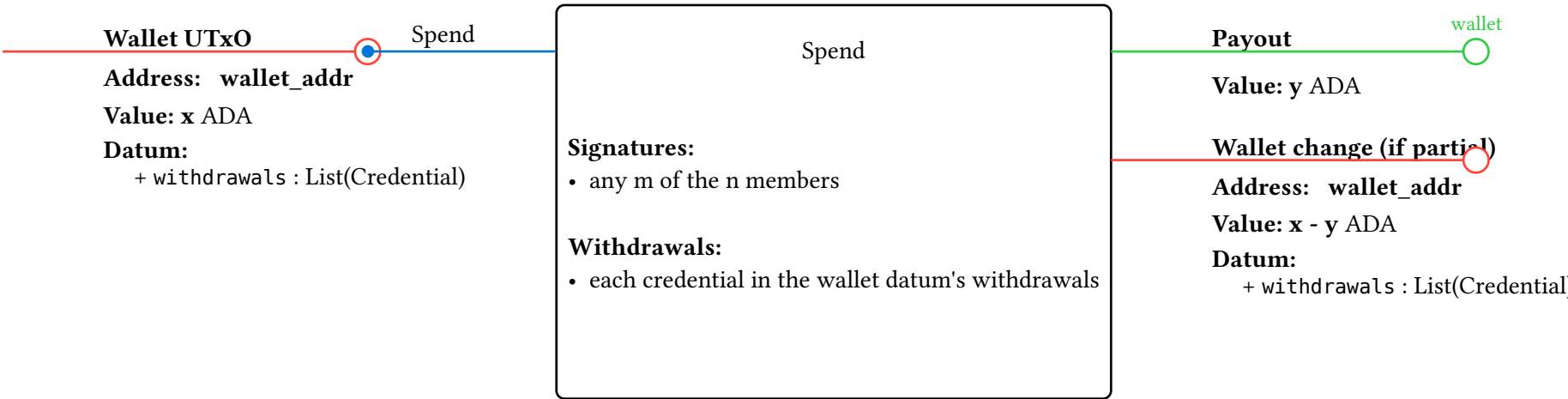
Contents

1 - Spend (base M-of-N + delegated authorizations) 2

2 - Update withdrawals (per-UTxO) 3

1 - Spend (base M-of-N + delegated authorizations)

The wallet pays out. The spending script enforces its base M-of-N signature floor and forwards every other check to the withdrawal scripts named in the wallet UTxO's datum; the single spend redeemer asserts their conjunction.



Notes:

- wallet_addr: payment is wallet_hash, staking any. Wallet UTxOs carry no datum; they may hold ada and arbitrary native assets.
- wallet_hash = hash(spend_validator(config, admin)): the spending script is parameterized by its configuration reference and the admin credential, so all wallet UTxOs share one address. Externalizing the members/threshold config lets it change without redeploying the script.
- base M-of-N floor: the wallet's configuration supplies members and threshold, and the spend requires at least threshold of members present in extra_signatories. A member may itself be a script credential, so the floor can be another multisig, a DAO, and so on. This design is agnostic of how that configuration is sourced.
- delegated authorizations: every Credential in the wallet UTxO's withdrawals datum field is a withdraw-0 staking script that must run in this transaction. The Spend redeemer asserts the conjunction — the M-of-N floor **and** all delegated checks — so no restriction can be silently omitted. The list is per-UTxO, so different wallet UTxOs may carry different restriction sets.
- the delegated scripts are the extension point: spending limits, time locks, recipient allow-lists, oracle attestations, ... each is an independent withdrawal script observed here, none known to the spending script.
- composability: the spend asserts only its own inputs and that the required withdrawals ran — never the total input or output counts, nor unrelated UTxOs.

Figure 1: Spend (base M-of-N + delegated authorizations)

2 - Update withdrawals (per-UTxO)

An authorized admin credential — a parameter of the spending script — rewrites a wallet UTxO’s delegated `withdrawals` list in place. Funds are untouched: same address, same value — only the restriction set changes.

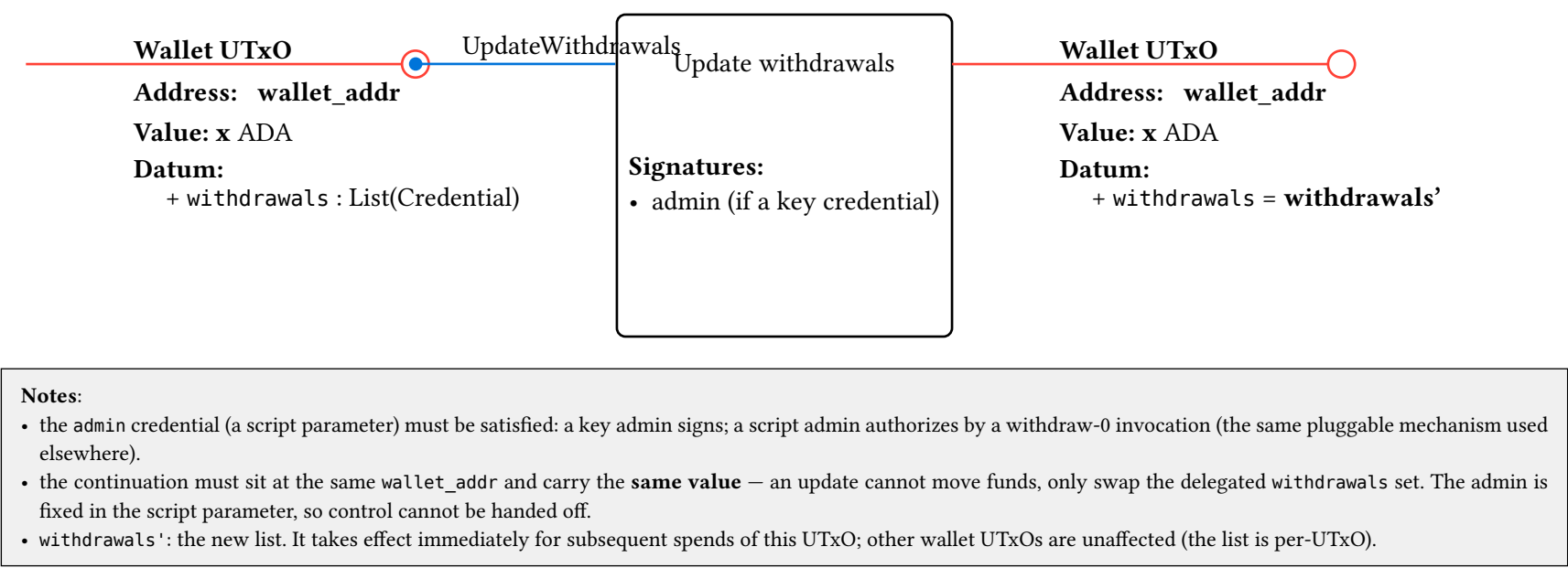


Figure 2: Update withdrawals (per-UTxO)